

1 (a) density = mass / volume B1 [1]

(b) density of liquids and solids same order as spacing similar / to about $2\times$
density of gases much less as spacing much more

or density of gases much lower hence spacing much more B1 [2]

(c) (i) density = $68 / [50 \times 600 \times 900 \times 10^{-9}]$ C1

= 2520 (allow 2500) kg m^{-3} A1 [2]

(ii) $P = F / A$ C1

= $68 \times 9.81 / [50 \times 600 \times 10^{-6}]$ C1

= $2.2 \times 10^4 \text{ Pa}$ A1 [3]

2 (a) torque is the product of one of the forces and the distance between forces M1